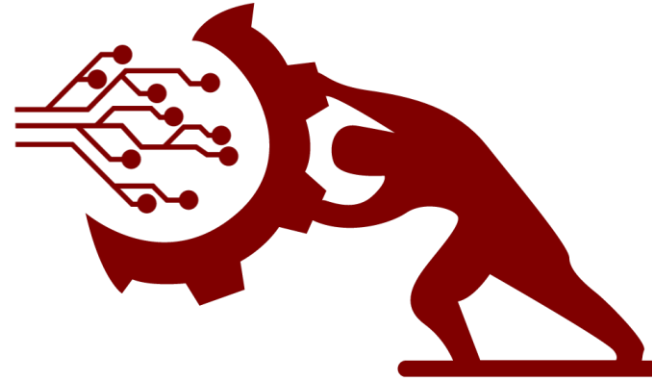


ENDEAVOUR



ENDEAVOUR ASSESSMENT 2025

Syllabus & Evaluation Scheme

<https://www.endeavourslit.in/>

Endeavour Assessment 2025 — Syllabus & Evaluation Scheme

Exam: Assessment Test — *UG & ICD*

Maximum Marks: 100

Duration: 90 minutes

(**note:** both UG and ICD contain same marking and evaluation scheme)

Exam Structure

- **Section A:** 24 MCQs × 1 mark = 24 marks (Physics, Chemistry, Mathematics, Aptitude). Negative marking: −0.25 per wrong answer.
- **Section B:** 13 MCQs × 2 marks = 26 marks (Physics, Chemistry, Mathematics, Aptitude). Negative marking: −0.25 per wrong answer.
- **Section C:** 10 Technical MCQs × 2 marks = 20 marks. No negative marking. Attempt all.
- **Section D:** 3 Descriptive questions = 30 marks. All compulsory. Questions cover project design, personal motivation, and logical reasoning.
- **Total:** 100 marks.

Evaluation Scheme

Section	Questions	Marks per Q	Section Marks	Negative Marking	Notes
Section 1	24	1	24	−0.25	Short recall MCQs
Section 2	13	2	26	−0.25	Conceptual and numerical MCQs
Section 3	10	2	20	None	Technical domain questions
Section 4	3	10	30	None	Descriptive, rubric-based
Total	50	—	100	—	—

Descriptive Rubrics

- **Q1 Project Design (10 marks):** innovation (3), technical correctness (3), clarity of plan (2), diagrams/presentation (2)
- **Q2 Motivation (10 marks):** alignment with goals (3), specific examples (3), growth plan (2), clarity of language (2)
- **Q3 Logic Puzzle (10 marks):** correct solution (4), reasoning steps (4), clarity and conciseness (2)

1. Syllabus for (UG1st 2nd & ICD 3rd year)

1.1. Modules

- **Module 1:** Physics Fundamentals — mechanics, E&M, optics, waves, modern physics
- **Module 2:** Chemistry Basics — adsorption, polymers, applied chemistry, electrochemistry
- **Module 3:** Mathematics Essentials — algebra, calculus, trigonometry, applied numerics
- **Module 4:** Aptitude & Reasoning — series, clock/calendar, logic puzzles
- **Module 5:** Technical Knowledge — sensors, Arduino, memory, communication, components
- **Module 6:** Descriptive Practice — project design, motivation writing, logic puzzle explanations

1.2. Subject Coverage

Subject	Subject Coverage	Number of questions
Physics	• Mechanics: free fall, momentum, torque, rotational equilibrium, simple harmonic motion • Electricity & Magnetism: circuits, induced emf, magnetic flux, capacitors (energy storage) • Optics: lenses, focal length, refraction through slabs, magnification • Waves: resonance, frequency–wavelength relation, ultrasonic sensor basics • Modern Physics: photoelectric effect (stopping potential, threshold frequency) • Measurement: dimensional analysis, unit consistency	4Questions (1 M) 3Questions (2 M)
Chemistry	• Physical Chemistry: adsorption, electrochemistry (redox, potential) • Organic Chemistry: polymerization, addition vs condensation • Applied Chemistry: soaps, detergents, emulsifiers, vitamins • Environmental Chemistry: lightning reactions, ozone formation, NO_x chemistry	4Questions (1 M) 3Questions (2 M)

Mathematics	- Algebra: quadratic equations, sequences and series - Calculus: limits, derivatives, integrals, continuity - Trigonometry: standard identities, inverse functions - Applied Numerics: pipes and cisterns, trains, mixtures, ages, ratios, percentages - Complex Numbers: polar representation, phasors	4Questions (1 M) 3Questions (2 M)
Aptitudes	- Calendar and clock problems - Series and pattern recognition, figure completion - Speed, distance, averages, ratios - Probability and simple statistics - Logic puzzles involving arrangement and selection	4Questions (1 M) 4Questions (2 M)
Technical Knowledge	- Sensors: Ultrasonic, PIR, LDR (principle and applications) - Arduino: Pinout, PWM, board types (Uno, Nano, Mega) - Memory: EEPROM basics - Communication: IR, Bluetooth, Wi-Fi, RFID - Actuators & Modules: servo, relay, buzzer, LCD/OLED displays - Circuits: breadboarding, wiring, current limits, safety practices	10 Questions (2M)

2. Syllabus for ICD (1st 2nd year)

2.1 Modules

- **Module 0:** Orientation & Exam Strategy — negative marking rules, time management
- **Module 1:** Physics Basics — electricity, optics, mechanics
- **Module 2:** Chemistry Basics — inorganic, organic, applied chemistry
- **Module 3:** Mathematics — algebra, trigonometry, geometry, probability, series
- **Module 4:** Aptitude — coding, series, calendars, trains, logic puzzles
- **Module 5:** Technical Knowledge — sensors, Arduino, EEPROM, communication
- **Module 6:** Descriptive Practice — project design, motivation writing, logic problem solving

2.2. Subject Coverage

Subject	Subject Coverage	Number of questions
Physics	• Electricity & Circuits: resistors in series/parallel, Ohm's law, current–voltage relations • Optics: mirrors, lenses, image formation • Magnetism: Fleming's rules, electromagnetic force • Mechanics: motion, force, simple applications	4Questions (1 M) 3Questions (2 M)
Chemistry	• General chemistry: biogas composition, acids in food (lactic acid in curd) • Inorganic: magnesium oxide layer, Plaster of Paris, metals and non-metals • Organic: alcohol functional groups, acetic acid formula • Applied: uses of gypsum, metallic properties	4Questions (1 M) 3Questions (2 M)
Mathematics	• Algebra: quadratic polynomials, discriminant conditions, linear equations • Geometry: distance formula, line division • Trigonometry: basic ratios, secant/tangent relations • Probability: valid ranges, simple problems • Series & patterns: number series, missing terms	4Questions (1 M) 3Questions (2 M)

Aptitudes	• Coding–decoding problems • Calendar and clock questions • Train and platform problems • Work, time, speed, ratio-based numericals • Figure series, analogy	4Questions (1 M) 4Questions (2 M)
Technical Knowledge	• Sensors: Ultrasonic, PIR, LDR (principle and applications) • Arduino: Pinout, PWM, board types (Uno, Nano, Mega) • Memory: EEPROM basics • Communication: IR, Bluetooth, Wi-Fi, RFID • Actuators & Modules: servo, relay, buzzer, LCD/OLED displays • Circuits: breadboarding, wiring, current limits, safety practices	10 Questions (2m)

#Instructor Notes

- Emphasize exam discipline (no electronic devices, no malpractice).
 - Encourage accuracy over guesswork in Sections A & B.
 - Urge full attempts in Section C since it has no penalties.
 - Provide clear practice rubrics for Section D responses.
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This syllabus and evaluation scheme outline provides UG & ICD students with a clear structure of subject coverage, section weightage, and preparation guidelines for the Endeavour Assessment 2025.

Study material for Technical section:

Introduction to Arduino Microcontroller

Overview of Arduino

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It is widely used for creating interactive projects and applications.

Arduino Boards and Their Differences

- **Arduino Uno:** Uses ATmega328P microcontroller, 14 digital IO pins (6 PWM), 6 analog inputs, 32 KB flash memory, 2 KB SRAM, 1 KB EEPROM.
- **Arduino Mega:** ATmega2560 microcontroller, 54 digital IO pins, 16 analog inputs, 256 KB flash memory.
- **Arduino Nano:** Smaller form factor using same ATmega328P chip, similar features to Uno but compact.
- Other boards include Arduino Due, Leonardo, Micro, etc., each serving different needs.

Board	Microcontroller	Flash Memory	Digital IO Pins	Analog Inputs	PWM Pins	SRAM	EEPROM
Arduino Uno	ATmega328P	32 KB	14	6	6	2 KB	1 KB
Arduino Mega	ATmega2560	256 KB	54	16	15	8 KB	4 KB
Arduino Nano	ATmega328P	32 KB	14	8	6	2 KB	1 KB

Digital and Analog Pins

- Digital pins can be set as INPUT or OUTPUT.
- Analog pins read continuous values (0-1023) via ADC (Analog to Digital Converter).
- PWM (Pulse Width Modulation) pins simulate analog output by switching ON/OFF rapidly with varying duty cycles.

Electrical Characteristics

- Operating voltage usually 5V (some boards support 3.3V).
- Maximum current per IO pin is 40 mA; exceeding this can damage the microcontroller.
- Logic HIGH means 5V on Arduino Uno, LOW means 0V.

Memory Types on Arduino

- **Flash Memory:** Stores the sketch (program).
- **SRAM:** Stores variables during execution.
- **EEPROM:** Stores data permanently across power cycles.

Basic Arduino Function Commands

- **pinMode(pin, mode):** Configures a pin as INPUT or OUTPUT.
- **digitalWrite(pin, value):** Outputs HIGH or LOW on a digital pin.
- **digitalRead(pin):** Reads the value from a digital pin.
- **analogRead(pin):** Reads analog input value.
- **analogWrite(pin, value):** Sends a PWM signal with duty cycle corresponding to value (0-25)

Basic Sensors

Ultrasonic Sensor

- Measures distance by sending and receiving sound waves.
- Calculates distance based on the time taken for echo to return.
- Used in robotic obstacle detection.

PIR Sensor

- Detects motion by sensing infrared radiation.
- Commonly used in security systems and automation.

LDR (Light Dependent Resistor)

- Resistor with resistance varying according to light intensity.
- Used for automatic lighting and detecting ambient light.

Wireless Modules

IR (Infrared) Communication

- Uses IR light to transmit data wirelessly.
- Common in remote controls.

Bluetooth Communication

- Wireless short-range communication.
- Suitable for wireless data exchange with smartphones.

RFID (Radio Frequency Identification)

- Uses radio waves to identify tags.
- Used in access control and inventory management.